

MD QUTUB UDDIN SAJIB, Ph.D.

Curriculum Vitae

✉ qsajib71@gmail.com

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Professional Profile

Environmental scientist with a Ph.D. in Environmental Science and Engineering and research experience in remote sensing, land surface temperature modeling, GIS, and environmental data analysis. Following completion of my doctorate, I spent six years at Mathematical Sciences Publishers, contributing to scholarly publishing workflows, software development, and $\text{\LaTeX}$ -based production systems. My research and professional interests include environmental data science, geospatial analysis, open-source scientific software, digital typography, and scholarly publishing.

Research Interests

- Environmental data science
- Remote sensing and geospatial analysis
- Environmental modeling
- Scientific computing
- Open-source scientific software
- Scholarly publishing, digital typography

Education

- Sep. 2014–Jun. 2020 **Doctor of Philosophy (Ph.D.) in Environmental Science and Engineering**
China University of Geosciences, Wuhan, China
Dissertation *Comparative Study of Land Surface Temperature from Landsat 8 in Chandina, Bangladesh*
Advisor **Dr. Tao Wang**, Professor, School of Environmental Studies
- Jul. 2011–Jun. 2012 **Master of Science (M.S.) in Environmental Science**
University of Dhaka, Dhaka, Bangladesh
Thesis *Identification and Assessment of Land Use and Land Cover Changes Using Remote Sensing and GIS Techniques at Kaaultia Union of Gazipur, Bangladesh*
Advisor **Dr. A.S.M. Mohiuddin**, Professor, Department of Soil, Water, and Environment
- Jul. 2004–Jun. 2010 **Bachelor of Science with Honors**
University of Dhaka, Dhaka, Bangladesh
Project *Green Technology*
Advisor **Dr. M. Shafiqur Rahman**, Professor, Department of Soil, Water, and Environment

Research Experience

- Sep. 2014–Jun. 2020 **Doctoral Research**
China University of Geosciences, Wuhan, China
Conducted doctoral research in remote sensing and environmental modeling, focusing on land surface temperature (LST) estimation from Landsat 8 imagery.
- Performed remote sensing analysis of Landsat 8 imagery using ENVI to estimate LST.
 - Processed vector data and conducted spatial analysis using ArcGIS.
 - Compared four LST estimation algorithms and evaluated their performance.
 - Developed well-documented, reusable R scripts for statistical analysis, comparison of LST algorithms, and publication-quality visualization; included representative code examples in the dissertation to promote reproducible research.
 - Published the research in a peer-reviewed journal.

Jul. 2011–Jun. 2012 **Master's Research**

University of Dhaka, Dhaka, Bangladesh

Conducted master's research on land-use and land-cover (LULC) change using remote sensing and GIS techniques.

- Investigated LULC change using multi-temporal satellite imagery.
- Performed spatial analysis and thematic mapping using GIS.
- Evaluated environmental changes through satellite image interpretation.

Professional Experience

Jun. 2020–Jun. 2026 **Production Editor**

Mathematical Sciences Publishers (MSP), Berkeley, California (Remote)

- Contributed to scholarly publishing through technical editing, production, and workflow optimization.
- Copyedited research articles across multiple mathematical disciplines for publication.
- Developed enhancements to $\text{\LaTeX}$ journal class files to improve publication workflows.
- Identified software issues in production scripts and provided technical feedback that contributed to workflow improvements.
- Collaborated with the production team to improve the journal publication system.

Supervisors **Silvio Levy** (2020–2025); **John Bourke** (2025–2026)

Jun. 2012–Aug. 2014 **Assistant Coordinator (Publication)**

Chhayanaut, Dhaka (On-site)

- Managed production of the quarterly *Bangladesher Hridoy Hotey*.
- Established and managed an in-house production workflow, replacing the previous outsourced process.
- Liaised with authors, coordinated proofs, and ensured delivery of final printed copies.
- Coordinated editorial schedules and workflows.
- Organized monthly editorial board meetings.

Supervisor **Prof. Dr. Sanjida Khatun**

Publications

Journal Article

- 2020 **M. Q. U. Sajib** and T. Wang. Estimation of land surface temperature in an agricultural region of Bangladesh from Landsat 8: intercomparison of four algorithms. *Sensors* 20(6), art. id. 1778. DOI 10.3390/s20061778.

Peer-Reviewed Conference Proceedings

- 2013 M. S. U. Rajib, L. Fan, **M. Q. U. Sajib**, and N. Z. Mou. Production and preferences to compatible solution: fuzzy evaluation of power sector in Bangladesh. In *Information Technology for Manufacturing Systems IV*, vol. 421 of *Applied Mechanics and Materials*, pp. 904–909. Trans Tech Publ. DOI 10.4028/www.scientific.net/AMM.421.904.
- 2011 M. S. U. Rajib, M. S. Alam, **M. Q. U. Sajib**, and N. Ahmed. Impact of relationship among the fundamental blocks in production process to implement re-engineering: an empirical analysis in SMEs. In *Machinery, Materials Science and Engineering Applications, MMSE2011*, vol. 228–229 of *Advanced Materials Research*, pp. 696–701. Trans Tech Publ. DOI 10.4028/www.scientific.net/AMR.228-229.696.
- 2011 **M. Q. U. Sajib**, M. S. U. Rajib, and M. S. Alam. An assessment on the adaptation of ISO 14000 in the fertilizer industry of Bangladesh for the sustainable development. In *Proceedings of the 2011 International Conference on Energy and Environmental Science (ICEES 2011)*, vol. 11 of *Energy Procedia*, pp. 2480–2488. Elsevier. DOI 10.1016/j.egypro.2011.10.568.

Book Chapters

- 2019 M. S. U. Rajib and **M. Q. U. Sajib**. Corporate disclosure measurement. In M. Khosrow-Pour, editor, *Advanced Methodologies and Technologies in Network Architecture, Mobile Computing, and Data Analytics*, chap. 36, pp. 489–501. IGI Global. DOI 10.4018/978-1-5225-7598-6.ch036.
- 2018 M. S. U. Rajib and **M. Q. U. Sajib**. Corporate disclosure measurement. In M. Khosrow-Pour, editor, *Encyclopedia of Information Science and Technology*, chap. 165, pp. 1896–1906. IGI Global, 4th ed. DOI 10.4018/978-1-5225-2255-3.ch165.
- 2017 M. S. U. Rajib, **M. Q. U. Sajib**, and M. Hoque. Strengthening the public sector accounting through ICT: the experience of a developing country. In H. Kaur, E. Lechman, and A. Marszk, editors, *Catalyzing Development through ICT Adoption: The Developing World Experience*, pp. 71–87. Springer. DOI 10.1007/978-3-319-56523-1_6.

TUGboat Article

- 2019 **M. Q. U. Sajib**. Typesetting the Bangla script in Unicode TeX engines — experiences and insights. *TUGboat* 40(3), pp. 263–269. URL tug.org/TUGboat/tb40-3/tb126sajib-bangla.pdf.

Skills

Programming	Python, Perl
Computing	Linux, Emacs
Statistical, Geospatial	R, QGIS, ArcGIS, ENVI, ERDAS Imagine
Digital Typography	T _E X, L _A T _E X, X _Y L _A T _E X, LuaL _A T _E X
Scientific Publishing	Journal production, workflow development, technical copyediting

Technical Contributions

Open-Source	Development of open-source tools for high-quality typesetting of the Bangla script with X _Y L _A T _E X and LuaL _A T _E X.
Production	Contributed to journal production workflows and L _A T _E X class file improvements at MSP.
Text Rendering	Investigated Bangla text rendering in Unicode T _E X engines, contributing to technical discussions with HarfBuzz maintainers and T _E X developers.
BanglaFineType	Founder and developer of BanglaFineType, an open initiative focused on the fine points of Bangla typography through research, engineering, education, and scholarly publishing.

Community Engagement

Gunijan

Jun. 2011–Aug. 2013 **Researcher and Writer**

Conducted biographical research and authored short biographies in Bangla of seven eminent personalities of Bangladesh for the Gunijan digital archive.

University of Dhaka

Jul. 2005–Jan. 2006 **Student Assistant**

Served as a student assistant at the Science Library, University of Dhaka.

References

Academic and professional references are available upon request.